

***Revised
Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AD1-097***

“Linwood 230 kV II”

67 MWE, 34.8 MWC Uprate

***October 2020
Revision 2***

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

Revisions from System Impact Study Issued December 2019:

The AD1-097 System Impact Study issued in December 2019 has been revised to incorporate the following changes:

Load flow analysis was retooled considering the project withdrawals and generator deactivations. Analysis shows that this project no longer contributes to the overload of the Delco Tap-Mickleton 230 kV Line. The Network Upgrade, n5802, reinforcement is no longer applicable to this project. References to the reinforcement have been deleted from the report.

Revisions from Original System Impact Study Issued October 2018:

The System Impact Study issued on October 19, 2018 inadvertently missed the summer load flow analysis results. The AD1-097 System Impact Study has been revised to incorporate the following changes:

Load flow analysis was retooled considering the project withdrawals and generator deactivations. Analysis shows that this project still contributes to an overload of the Delco Tap-Mickleton 230 kV Line. The Network Upgrade Number to reinforce the line is n5802. The reinforcement includes replacement of substation equipment and substation bus at the Mickleton Substation (Atlantic City Electric “ACE” zone).

Though there is no cost allocation for AD1-097 at this time given cost allocation rules, there is the possibility that this project could have cost allocation in the future if prior queue projects withdraw. This project needs to have this reinforcement of the Delco Tap-Mickleton 230 kV line completed prior to going into service. Therefore, a Facilities Study is required to determine the

engineering level scope, cost and estimate of the reinforcement. ACE's initial estimate is 18 months to complete the upgrade.

Changes from Feasibility Study to Impact Study Phase:

Per request of the Interconnection Customer, the increase in Maximum Facility Output (MFO)/Energy was reduced from 88 MW to 67 MW and the increase in Capacity was reduced from 109.3 MW to 34.8 MW for the AD1-097 project. PJM determined this change was not a material modification. See the updated values in the table below.

AD1-097	Original Uprate Request (MW)	New Uprate Request (MW)
MFO	88	67
Capacity	109.3	34.8

General

Marcus Hook Energy, L.P., the Interconnection Customer (IC), has proposed an increase in output of their existing Marcus Hook (Phillips Island) natural gas generating facility located at 100 Green Street, Marcus Hook, Pennsylvania. The uprate will increase the maximum facility output (MFO) of the facility by **67 MW**, for a total plant MFO of 919 MW. The uprate will also increase the Capacity value by **34.8 MW**, for a total of 847.8 MW being recognized by PJM as Capacity. (See the summary tables below.) The Commercial Operation Date proposed for this project is **May 1, 2020**.

Queue Number	MFO (MW) and Winter Energy	Capacity (MW)
Existing Facility	852	813
AD1-097	+ 67 MW increase	+34.8 MW increase
Total	919	847.8

PJM studied the proposed request and evaluated it for compliance with reliability criteria for summer peak conditions in 2021.

Point of Interconnection

AD1-097 “Linwood 230 kV II” uprate project will be at increase to the existing Marcus Hook (Phillips Island) natural gas generation plant which interconnects to the PECO transmission system at the Linwood Substation 230 kV bus.

Attachment 1 shows the one line diagram of the project and the existing point of interconnection with the Linwood 230 kV substation. As indicated, it was studied as a 919 MW injection to the PECO 230 kV system at the existing 230 kV connection for the Marcus Hook generation plant at PECO’s Linwood substation. The PECO facilities required to be upgraded for the primary direct connection, non-direct connection, and network upgrades of the generation project and the associated cost estimate are outlined below.

Cost Summary

The AD1-097 “Linwood 230 kV II” project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 0
Direct Connection Network Upgrades	\$ 0
Non Direct Connection Network Upgrades	\$ 15,000
New System Upgrades	\$ 0
Contribution to Previously Identified Upgrades	\$ 0
Total Costs	\$ 15,000

The transmission and substation costs given above exclude any applicable state or federal taxes. If at a future date Federal CIAC (contribution in aid of construction) taxes are deemed necessary by the IRS for this project, PECO shall be reimbursed by the Interconnection Customer for such taxes.

The required Attachment Facilities, Direct Connection, and Non-Direct Connection work for the interconnection of the AD1-097 generation uprate project to the PECO Transmission System is detailed in the following sections. The associated one-line with the generation project is shown in Attachment 1.

Attachment Facilities

There is no Attachment Facilities work required for this project.

Direct Connection Cost Estimate

There is no Direct Connection scope of work required for this project.

Non-Direct Connection Cost Estimate

The total preliminary cost estimate for the Non-Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Protection System changes at Linwood Substation (PECO) (PJM Network Upgrade Number n5778)	\$ 15,000
Total Non-Direct Connection Facility Costs	\$ 15,000

The total Non-Direct Connection cost estimate for the AD1-097 project is approximately **\$15,000**. This work is to review the protection system settings at PECO's Linwood Substation and to implement any necessary changes.

Schedule

Based on the extent of the PECO primary Non-Direct Connection work required to support the AD1-097 generation project, it is expected to take a **minimum of 4 months** from the date of a fully executed Interconnection Construction Service Agreement to complete the work required to accommodate the customer's request.

PECO Interconnection Requirements

The Interconnection Customer facilities must be designed in accordance with the "Exelon Utilities Transmission Facility Interconnection Requirements document dated January 1, 2019. This document is located at the following link:

<http://www.pjm.com/~media/planning/plan-standards/private-ce/exelon-utilities-transmission-facility-interconnection-requirements.ashx>

Revenue Metering and SCADA Requirements

PJM Requirements

Existing metering is adequate for this request.

Network Impacts

The Queue Project AD1-097 was evaluated as a 67 MW MFO (Capacity 34.8 MW) uprate injection into the Linwood 230 kV bus. Project AD1-097 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AD1-097 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Base Case Used

Summer Peak Analysis – 2021 Case

Contingency Descriptions

None

Summer Peak Analysis - 2021

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

None

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None

Steady-State Voltage Requirements

(Results of the steady-state voltage studies should be inserted here)

None

Stability and Reactive Power Requirement for Low Voltage Ride Through

(Summary of the VAR requirements based upon the results of the dynamic studies)

No mitigations were found to be required. (See Attachment 2 for Stability Study Report.)

Short Circuit

(Summary of impacted circuit breakers)

None

Affected System Analysis & Mitigation

NYISO Impacts:

NYISO Impacts to be determined during later study phases (as applicable).

Delivery of Energy Portion of Interconnection Request

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Only the most severely overloaded conditions are listed. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed, which will study all overload conditions associated with the overloaded element(s) identified.

None

Light Load Analysis - 2021

No light load thermal impacts.

System Reinforcements

Short Circuit

(Summary form of Cost allocation for breakers will be inserted here if any)

None

Stability and Reactive Power Requirement

(Results of the dynamic studies should be inserted here)

None

Summer Peak Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

None

Light Load Load Flow Analysis Reinforcements

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None

Contribution to Previously Identified System Reinforcements

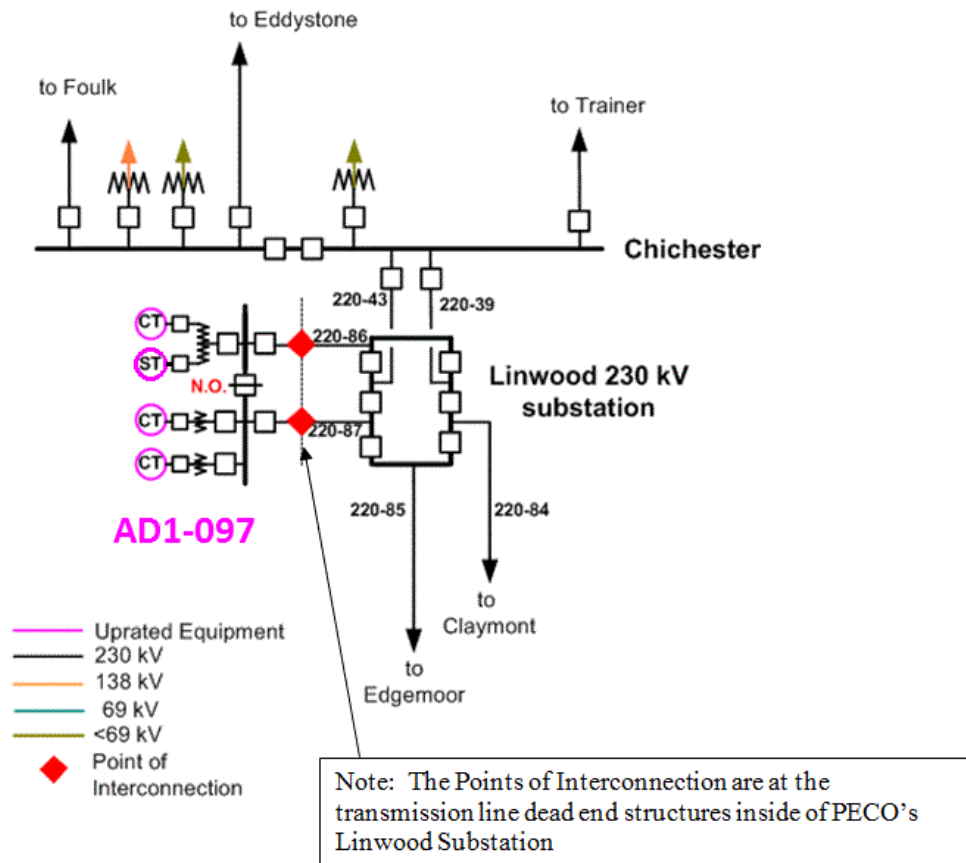
(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

(Summary form of Cost allocation for transmission lines and transformers will be inserted here if any)

None

=====

Attachment 1: (AD1-097) Generation Project Planning Interconnection Single Line Diagram



Attachment 2: (AD1-097) Generation Project
Dynamic Stability Analysis Results

AD1-097
System Impact Study

Dynamic Simulation Analysis

Version 0

March 6, 2018

TABLE OF CONTENTS

Executive Summary.....3

1. Introduction5

2. Description of Project.....6

3. Loadflow and Dynamics Case Setup11

4. Fault Cases.....12

5. Evaluation Criteria13

6. Summary of Results14

7. System Reinforcement Requirements15

Executive Summary

Generator Interconnection Request AD1-097 project is for a 67 MW Maximum Facility Output (MFO) uprate to the existing Phillips Island generating facility with the Point of Interconnection (POI) on Linwood 230kV station in PECO system located in Marcus Hook, Pennsylvania.

This report describes a dynamic simulation analysis of AD1-097 as part of the overall system impact study.

The load flow scenario for the analysis was based on the RTEP 2021 Light Load case, modified to include applicable queue projects. AD1-097 has been dispatched online at maximum power output, with regulating Linwood 230kV bus voltage at 1.004 p.u., which is consistent with the default generator reference voltage specified in PJM Manual 03 Transmission Operations Section 3.3.3 for generator connections to the PJM 230 kV system.

The AD1-097 queue project was tested for compliance with NERC, PJM and other applicable criteria. The range of contingencies evaluated was limited to that necessary to assess compliance and each was limited to a 20-second simulation time period.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

The system was tested for a system intact condition and the fault types listed above. Specific fault descriptions and breaker clearing times used for this study are provided in the result table.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For the fault contingencies tested on the 2021 light load case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AD1-097 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

No mitigations were found to be required.

1. Introduction

Generator Interconnection Request AD1-097 project is for a 67 MW Maximum Facility Output (MFO) uprate to the existing Phillips Island generating facility with the Point of Interconnection (POI) on Linwood 230kV station in PECO system located in Marcus Hook, Pennsylvania.

This analysis is effectively a screening study to determine whether the addition of AD1-097 will meet the dynamic requirements of the NERC, PJM and Transmission Owner reliability standards.

In this report the AD1-097 project and how it is proposed to be connected to the grid are first described, followed by a description of how the project is modeled in this study. The fault cases are then described and analyzed, and lastly a discussion of the results is provided.

2. Description of Project

Generator Interconnection Request AD1-097 project is for a 67 MW Maximum Facility Output (MFO) uprate to the existing Phillips Island generating facility with the Point of Interconnection (POI) on Linwood 230kV station in PECO system located in Marcus Hook, Pennsylvania.

Figure 1 shows the simplified one-line diagram of the AD1-097 loadflow model. Table 1 lists the parameters given in the impact study data and the corresponding parameters of the AD1-097 loadflow model.

The dynamic models for AD1-097 plant are based on the PSS\E standard model with parameters supplied by the developer.

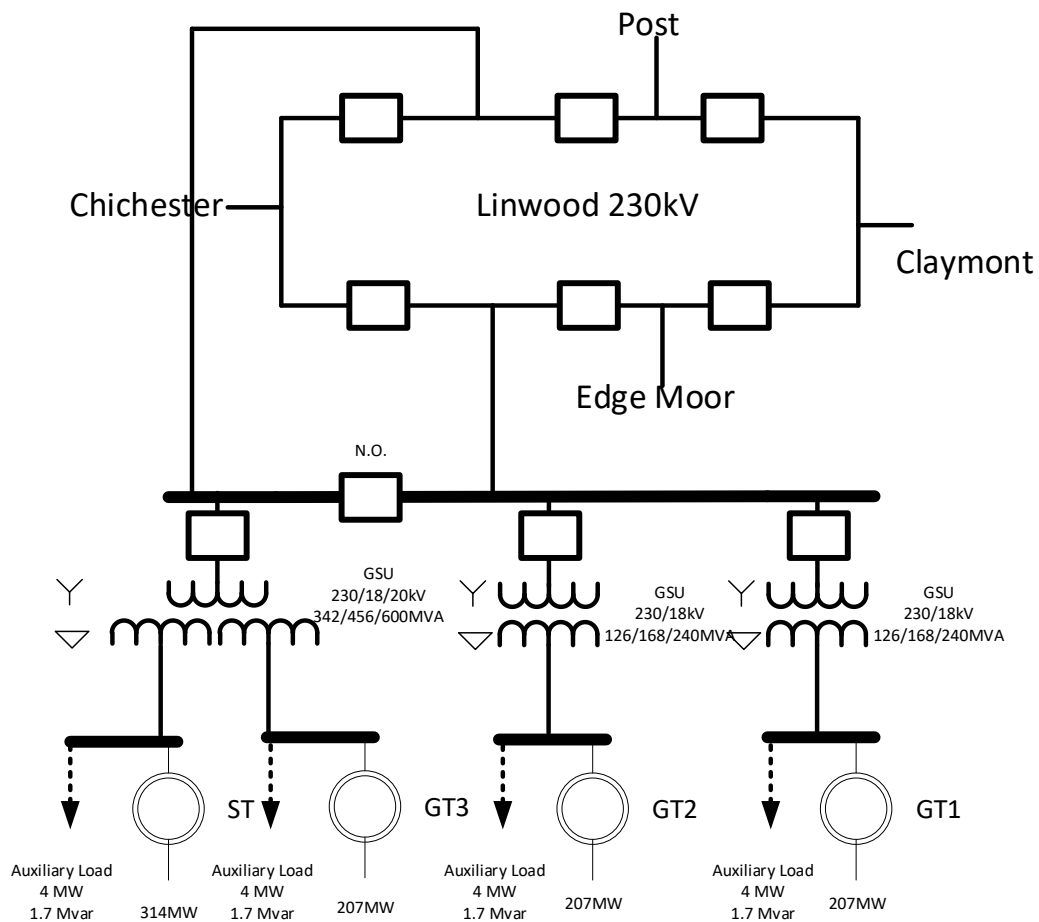


Figure 1: AD1-097 Plant Model

Table 1: AD1-097 Plant Model

	Impact Study Data	Model
Generators	3 × 207 MW GE GT MVA base = 221.4 MVA Vt = 18 kV Unsaturated sub-transient reactance = 0.185 pu @ MVA base	3 × 207 MW GE GT Pgen 207 MW Pmax 207 MW Pmin 116 MW Qmax 120 MVA_r Qmin -95 MVA_r Mbase 221.4 MVA Zsorce j0.185 pu @ Mbase
	1 × 314 MW TOSHIBA ST MVA base = 325 MVA Vt = 20 kV Unsaturated sub-transient reactance = j0.242 pu @ MVA base	1 × 314 MW TOSHIBA ST Pgen 314 MW Pmax 314 MW Pmin 59 MW Qmax 100 MVA_r Qmin -100 MVA_r Mbase 325 MVA Zsorce j0.242 pu @ Mbase

	Impact Study Data	Model
GSU transformer(s)	GT1 GSU 230/18 kV two winding transformer	GT1 GSU 230/18 kV two winding transformer
	Rating = 126/168/240 MVA	Rating = 126/168/240 MVA
	Transformer base = 126 MVA	Transformer base = 126 MVA
	Impedance = 0.0023 + j0.0867pu @ MVA base	Impedance = 0.0023 + j0.0867pu @ MVA base
	Number of taps = 5	Number of taps = 5
	Tap step size = 2.5%	Tap step size = 2.5%
	GT2 GSU 230/18 kV two winding transformer	GT2 GSU 230/18 kV two winding transformer
	Rating = 126/168/240 MVA	Rating = 126/168/240 MVA
	Transformer base = 126 MVA	Transformer base = 126 MVA
	Impedance = 0.0023 + j0.0857pu @ MVA base	Impedance = 0.0023 + j0.0857pu @ MVA base
	Number of taps = 5	Number of taps = 5
	Tap step size = 2.5%	Tap step size = 2.5%
	GT3+ST GSU 230/18/23kV three winding transformer	GT3+ST GSU 230/18/23kV three winding transformer

	Impact Study Data	Model
Auxiliary load	4 x 4 MW, 1.7 MVar	4 x 4 MW, 1.7 MVar
Station load	N/A	N/A
Attachment line	1.15 Miles	R = 0.0001 p.u. X = 0.0007 p.u. B = 0.0023 p.u.

3. Loadflow and Dynamics Case Setup

The dynamics simulation analysis was carried out using PSS/E Version 33.7.

The load flow scenario and fault cases for this study are based on PJM's Regional Transmission Planning Process¹.

The selected load flow scenario is the RTEP 2021 light load case with the following modifications:

- a) Addition of all applicable queue projects prior to AD1-097.
- b) Addition of AD1-097 queue project.
- c) Removal of withdrawn and subsequent queue projects in the vicinity of AD1-097.
- d) Dispatch of units in the PJM system to maintain slack generators within limits.

The AD1-097 initial conditions are listed in Table 2, indicating maximum power output with regulating Linwood 230kV bus voltage at 1.004 p.u., which is consistent with the default generator reference voltage specified in PJM Manual 03 Transmission Operations Section 3.3.3 for generator connections to the PJM 230 kV system.

Table 2: AD1-097 machine initial conditions

Bus	Name	Unit	PGEN	QGEN	ETERM	POI Voltage
213888	PHLISCT1 18.000	1	207	-4	0.97	1.004
213889	PHLISCT2 18.000	1	207	-4	0.97	1.004
213890	PHLISCT3 18.000	1	207	-4	0.97	1.004
213893	PHLISST1 20.000	1	314	-4	0.97	1.004

Generation within the PJM500 system (area 225 in the PSS/E case) and within the vicinity of AD1-097 has been dispatched online at maximum output (P_{MAX}). The dispatch of generation in the vicinity of AD1-097 is given in Attachment 3.

¹ Manual 14B: PJM Region Transmission Planning Process, Rev 33, May 5 2016, Attachment G : PJM Stability, Short Circuit, and Special RTEP Practices and Procedures.

4. Fault Cases

Tables 3 listed the contingencies and results that were studied, with representative worst case total clearing times provided by PJM. Each contingency was studied over a 20 second simulation time interval.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

The system was tested for a system intact condition and the fault types listed above.

5. Evaluation Criteria

This study is focused on AD1-097, along with the rest of the PJM system, maintaining synchronism and having all states return to an acceptable new condition following the disturbance. The recovery criteria applicable to this study are as per PJM's Regional Transmission Planning Process and Transmission Owner criteria:

- a) The system with AD1-097 included is transiently stable and post-contingency oscillations should be positively damped with a damping margin of at least 3%
- b) The AD1-097 is able to ride through faults (except for faults where protective action trips AD1-097).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

6. Summary of Results

Plots from the dynamic simulations are provided in Attachment 4, with results summarized in Table 3.

Frequency protection in the model is disabled due to the deficiency of PSSE frequency calculation for inverter based generation facilities.

For the studied fault contingencies tested on the 2021 light load case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 3%.
- b) The AD1-097 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

7. System Reinforcement Requirements

None.

Table 3: Contingency List and Stability Study Results**P0: Stead State**

Fault ID	Duration
P0.00	Steady State 20 sec run

P1: Three Phase Faults with normal clearing

Fault ID	Fault description	Clearing Time Normal & HSR (Cycles)	Results
P1.00	3ph @ Linwood 230kV, normal clear loss of AD1-097 GT1 & GT2	5	Stable
P1.01	3ph @ Linwood 230kV, normal clear loss of AD1-097 GT3 & ST	5	Stable
P1.02	3ph @ Linwood – Chichester 230kV line, normal clear	5	Stable
P1.03	3ph @ Linwood – Post 230kV line, normal clear	5	Stable
P1.04	3ph @ Linwood – Claymont 230kV line, normal clear	7	Stable
P1.05	3ph @ Linwood – Edge Moor 230kV line, normal clear	7	Stable
P1.06	3ph @ Claymont – Linwood 230kV line, normal clear	7	Stable
P1.07	3ph @ Claymont – Edge Moor 230kV line, normal clear	7	Stable
P1.08	3ph @ Claymont 230/69kV TF 20, normal clear loss of Claymont 69kV bus	9	Stable
P1.09	3ph @ Edge Moor – Linwood 230kV line, normal clear	7	Stable
P1.10	3ph @ Edge Moor – Claymont 230kV line, normal clear	7	Stable
P1.11	3ph @ Edge Moor – Harmony 230kV line, normal clear	7	Stable
P1.12	3ph @ Edge Moor 230/138kV TF 20, normal clear	9	Stable
P1.13	3ph @ Edge Moor 230kV, normal clear loss of HR4	7	Stable
P1.14	3ph @ Edge Moor 230kV, normal clear loss of EM5	7	Stable
P1.15	3ph @ Post – Linwood 230kV line, normal clear	5	Stable
P1.16	3ph @ Post – Chichester 230kV line, normal clear	5	Stable
P1.17	3ph @ Post 230/13.8kV TF 2, normal clear	5	Stable
P1.18	3ph @ Post 230/13.8kV TF 3, normal clear	5	Stable
P1.19	3ph @ Chichester – Linwood 230kV line, normal clear	5	Stable

Fault ID	Fault description	Clearing Time Normal & HSR (Cycles)	Results
P1.20	3ph @ Chichester – Post 230kV line, normal clear	5	Stable
P1.21	3ph @ Chichester – Trainer 230kV line, normal clear	5	Stable
P1.22	3ph @ Chichester 230/13.8kV TF 5, normal clear	5	Stable
P1.23	3ph @ Chichester 230/13.8kV TF 3, normal clear	5	Stable
P1.24	3ph @ Chichester – Eddystone 230kV line, normal clear	5	Stable
P1.25	3ph @ Chichester – Foulk 230kV line, normal clear	5	Stable
P1.26	3ph @ Chichester 230kV, normal clear loss of Chichester 230/138kV TF 9A and 9B	8	Stable

P4: SLG Stuck Breaker (SB) Faults at Backup Clearing

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	Results
P4.01	SLG @ Linwood 230kV, normal clear loss of Linwood – Chichester 230kV, SB @ Linwood, delayed clear loss of AD1-097 GT1 & GT2	5/15	Stable
P4.02	SLG @ Linwood 230kV, normal clear loss of Linwood – Chichester 230kV, SB @ Linwood, delayed clear loss of AD1-097 GT3 & ST	5/15	Stable
P4.03	SLG @ Linwood 230kV, normal clear loss of Linwood – Post 230kV, SB @ Linwood, delayed clear loss of AD1-097 GT3 & ST	5/15	Stable
P4.04	SLG @ Linwood 230kV, normal clear loss of Linwood – Edge Moor 230kV, SB @ Linwood, delayed clear loss of AD1-097 GT1 & GT2	7/15	Stable
P4.05	SLG @ Linwood 230kV, normal clear loss of Linwood – Edge Moor 230kV, SB @ Linwood, delayed clear loss of Linwood – Claymont 230kV	7/17.5	Stable
P4.06	SLG @ Linwood 230kV, normal clear loss of Linwood – Post 230kV, SB @ Linwood, delayed clear loss of Linwood – Claymont 230kV	5/17.5	Stable
P4.07	SLG @ Claymont 230kV, normal clear loss of Claymont 230/69kV TF 20, SB @ Claymont, delayed clear loss of Claymont – Linwood 230kV line	9/17.5	Stable
P4.08	SLG @ Claymont 230kV, normal clear loss of Claymont 230/69kV TF 20, SB @ Claymont, delayed clear loss of Claymont – Edge Moor 230kV line	9/17.5	Stable

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	Results
P4.09	SLG @ Edge Moor 230kV, normal clear loss of Edge Moor – Claymont 230kV line, SB @ Edge Moor, delayed clear loss of Edge Moor 230/138kV TF 20	7/21	Stable
P4.10	SLG @ Edge Moor 230kV, normal clear loss of HR4, SB @ Edge Moor, delayed clear loss of Edge Moor 230/138kV TF 20	7/21	Stable
P4.11	SLG @ Edge Moor 230kV, normal clear loss of HR4, SB @ Edge Moor, delayed clear loss of EM5	7/17.5	Stable
P4.12	SLG @ Edge Moor 230kV, normal clear loss of HR4, SB @ Edge Moor, delayed clear loss of Edge Moor – Harmony 230kV line	7/17.5	Stable
P4.13	SLG @ Edge Moor 230kV, normal clear loss of EM5, SB @ Edge Moor, delayed clear loss of Edge Moor – Linwood 230kV line	7/17.5	Stable
P4.14	SLG @ Post 230kV, normal clear loss of Post 230/13.8kV TF 2, SB @ Post, delayed clear loss of Post – Chichester 230kV	5/15	Stable
P4.15	SLG @ Post 230kV, normal clear loss of Post 230/13.8kV TF 2, SB @ Post, delayed clear loss of Post – Linwood 230kV	5/15	Stable
P4.16	SLG @ Chichester 230kV, normal clear, SB @ Chichester, delayed clear loss of Chichester – Linwood 230kV, Chichester – Trainer 230kV, Chichester – Post 230kV, Chichester 230/13.8kV TF 5	5/15	Stable
P4.17	SLG @ Chichester 230kV, normal clear, SB @ Chichester, delayed clear loss of Chichester – Eddystone 230kV, Chichester – Foulk 230kV, Chichester 230/138kV TF 9A and 9B, Chichester 230/13.8kV TF 3	5/20	Stable

P5: SLG Fault with Delayed Clearing

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	Results
P5.01	SLG @ 80% of Linwood – Claymont 230kV line, relay failure @ Linwood, delayed clear	7/25	Stable
P5.02	SLG @ 80% of Claymont – Linwood 230kV line, relay failure @ Claymont, delayed clear	7/25	Stable
P5.03	SLG @ 80% of Linwood – Edge Moor 230kV line, relay failure @ Linwood, delayed clear	7/25	Stable
P5.04	SLG @ 80% of Edge Moor – Linwood 230kV line, relay failure @ Edge Moor, delayed clear	7/25	Stable
P5.05	SLG @ 80% of Claymont – Edge Moor 230kV line, relay failure @ Claymont, delayed clear	7/25	Stable
P5.06	SLG @ 80% of Edge Moor – Claymont 230kV line, relay failure @ Edge Moor, delayed clear	7/25	Stable

Fault ID	Fault description	Clearing Time Near & Remote (Cycles)	Results
P5.07	SLG @ 80% of Linwood – Post 230kV line, relay failure @ Linwood, delayed clear	5/30	Stable
P5.08	SLG @ 80% of Post – Linwood 230kV line, relay failure @ Post, delayed clear	5/30	Stable
P5.09	SLG @ 80% of Linwood – Chichester 230kV line, relay failure @ Linwood, delayed clear	5/30	Stable
P5.10	SLG @ 80% of Chichester – Linwood 230kV line, relay failure @ Chichester, delayed clear	5/30	Stable
P5.11	SLG @ 80% of Chichester – Post 230kV line, relay failure @ Chichester, delayed clear	5/30	Stable
P5.12	SLG @ 80% of Post – Chichester 230kV line, relay failure @ Post, delayed clear	5/30	Stable

P7: SLG Fault with Common Structure Failure

Fault ID	Fault description	Clearing Time (Cycles)	Results
P7.01	CONTINGENCY 'DPL_P7_1_B47_DPL1'	7	Stable
P7.02	CONTINGENCY 'DPL_P7_1_B47_DPL2'	7	Stable
P7.03	CONTINGENCY 'DPL_P7_1_DBL_1NCB_A' /* #1 & #2 KEENEY-STEELE 230	7	Stable
P7.04	CONTINGENCY 'DPL_P7_1_DBL_1NCB_B' /* #1 & #2 KEENEY-STEELE 230	7	Stable
P7.05	CONTINGENCY 'DPL_P7_1_DBL_3NC' /* HARMNY-KEENY 230;KEENY-CHRYSLR 230	7	Stable
P7.06	CONTINGENCY 'DPL_P7_1_DBL_5NC' /* EDGE MOOR-CLAYMONT 230;EDGE MOOR-LINWOOD(PECO) 230	7	Stable
P7.07	CONTINGENCY 'DPL_P7_1_DBL_6NC' /* EDGE MOOR-LINWOOD(PECO) 230;CLAYMONT-LINWOOD(PECO) 230	7	Stable